

Jennifer Lu

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EDUCATION

Purdue University

August 2024 - December 2027

Computer Science/ML concentration, Mathematics, Artificial Intelligence B.S.

West Lafayette, IN

- Dean's List all semesters — GPA: 3.8/4.0
- Relevant Coursework: Data Structures, Programming in C, Linear Algebra II, Probability, Differential Equations

EXPERIENCE

Data Science Intern

May 2026 – August 2026

Pacific Gas & Electric Company

Oakland, CA

- Built NLP classification models to independently predict incident attributes from 2.5K event summaries, automating data intake and reducing incident dataset population time from 20 days to 4 hours.
- Developed text classification pipelines using TF-IDF with Logistic Regression; addressed severe class imbalance with SMOTE, achieved >0.80 F1 across all five classification tasks.
- Optimized incident-attribute classifiers through K-fold cross-validation, tuning TF-IDF and Logistic Regression parameters to improve performance across varying class distributions.

Undergraduate Researcher

January 2026 – May 2026

Johnson & Johnson - Purdue Data Mine

West Lafayette, IN

- Engineered an interactive RShiny clinical-trial analytics dashboard for Johnson & Johnson, integrating natural-language querying, dynamic tables, and automated visualizations
- Built PostgreSQL-backed data workflows to retrieve and analyze patient-related data, supporting interactive exploration of clinical trends and characteristics.
- Contributed to a natural-language querying pipeline using an LLM for schema-aware SQL generation, containerized with Docker for reproducible deployment.

Undergraduate Researcher

August 2025 – December 2025

Autonomous Motorsports Purdue

West Lafayette, IN

- Fine-tuned a TwinLiteNetPlus semantic segmentation model (pretrained on BDD100K) using custom and synthetic AMP karting datasets to improve drivable-area and cone detection.
- Built a synthetic data generation pipeline with augmentation and randomized object placement, expanding training data with 3k+ kart and 6k+ cone images.

AI Intern

June 2025 – August 2025

AI Line of Sight

Remote

- Developed a C# Model Context Protocol (MCP) server for the BizFirstAI product to prototype AI-driven functionality and support future product integrations.

PROJECTS

Project SPR-4622: Integration of BIM for Bridge Modeling | *AutoIC Lab*

January 2025 – May 2025

- Built a Python-based tool with OpenCV to extract and digitize structural data from bridge plan PDF documents
- Fine-tuned RoBERTa based model on domain-specific bridge terminology to enhance OCR accuracy

Prison Assault Prediction Model | *Management Performance Hub*

January 2025 – May 2025

- Developed a classification model to predict prison assault incidents from facility-level data for the Management Performance Hub, achieving 84% accuracy with K-Nearest Neighbors.
- Benchmarked K-Nearest Neighbors against Random Forest and XGBoost to evaluate alternative classification approaches and select the highest-performing model.

TECHNICAL SKILLS

Programming Languages: Python, Java, R, C, C++, SQL

Libraries/Tools: PyTorch, OpenCV, scikit-learn, Pandas, Docker, PostgreSQL, RShiny

Languages: English, Chinese